

Vector spaces

VECTOR SPACE

Let V be a nonempty set with two operations vector addition and scalar multiplication is said to be a **vector space**(or) **linear space** over a field F , if for all $x, y \in V$ and $a, b \in F$ the following conditions hold

1. $(V, +)$ is an abelian group
2. $(ab)x = a(bx)$, for each pair of element $a, b \in F$ and $x \in V$
3. $a(x + y) = ax + ay$, for each element $a \in F$ and for each pair of elements $x, y \in V$
4. $(a + b)x = ax + bx$, for each element $x \in V$ and for each pair elements $a, b \in F$
5. $1 \cdot x = x$, for each element $x \in V$.

Note

The elements of the field F are called **scalars** and the elements of the vector space V are called **vectors**

Definition. A non-empty set V is said to be a **vector space** (or) **linear space** over a field F , if in V there are defined two operations
vector addition: For each pair of element $x, y \in V$ there is a unique element $x + y \in V$ and
scalar multiplication: For each element $x \in V$ and for each $a \in F$ there is a unique element $ax \in V$ satisfying the following:

(VS 1) $x + y = y + x, \forall x, y \in V$

(VS 2) $(x + y) + z = x + (y + z), \forall x, y, z \in V$

(VS 3) there exists an element 0 in V such that $x + 0 = 0 + x = x$,
for each $x \in V$

(VS 4) for each $x \in V$, there exists $y \in V$ such that
 $x + y = y + x = 0$

(VS 5) for each element, $x \in V$, $1x = x$

(VS 6) for each pair of element $a, b \in F$ and $(ab)x = a(bx)$

(VS 7) for each element $a \in F$ and for each pair of elements
 $x, y \in V$, $a(x + y) = ax + ay$

(VS 8) for each element $x \in V$ and for each pair elements $a, b \in F$,
 $(a + b)x = ax + bx$

Example 1

Check $V = \{(x_1, x_2) \in \mathbb{R}^2 : x_1, x_2 \in \mathbb{R}\}$ is a vector space over $\mathbb{R}$ or not, under two operations define

vector addition:

$$x + y = (x_1 + y_1, x_2 + y_2) \quad \forall x = (x_1, x_2), y = (y_1, y_2) \in \mathbb{R}^2$$

scalar multiplication: $ax = (ax_1, ax_2) \quad \forall x = (x_1, x_2) \in \mathbb{R}^2 \text{ and } a \in \mathbb{R}$

Solution:

Given that $V = (\mathbb{R}^2, +, \cdot)$ and $F = \mathbb{R}$.

To prove: V is an abelian group

Closure Axiom: Let $x, y \in \mathbb{R}^2$. Then

$$\begin{aligned} x + y &= (x_1, x_2) + (y_1, y_2) \\ &= (x_1 + y_1, x_2 + y_2) \in \mathbb{R}^2 \end{aligned}$$

Thus $x + y \in \mathbb{R}^2$. Hence closure property is satisfied.

Associative Property: Let $x, y, z \in R^2$. Then

$$\begin{aligned}x + (y + z) &= (x_1, x_2) + [(y_1, y_2) + (z_1, z_2)] \\&= (x_1, x_2) + [(y_1 + z_1, y_2 + z_2)] \\&= (x_1 + y_1 + z_1, x_2 + y_2 + z_2) \\&= (x + y) + z\end{aligned}$$

Hence associative property is satisfied.

Existence of Identity: Let $e = (e_1, e_2)$, where $e_1, e_2 \in R$. Then

$$\begin{aligned}x + e &= x \\(x_1, x_2) + (e_1, e_2) &= (x_1, x_2) \\(e_1, e_2) &= (x_1, x_2) - (x_1, x_2) \\&= (0, 0) \in R^2\end{aligned}$$

Hence identity exists.

Existence of Inverse: Let y be the inverse of x .
Then

$$\begin{aligned}x + y &= e \\(x_1, x_2) + (y_1, y_2) &= (e_1, e_2) \\(y_1, y_2) &= (e_1, e_2) - (x_1, x_2) \\&= (0, 0) - (x_1, x_2) \\&= (0, 0) + (-x_1, -x_2) \\&= (0 - x_1, 0 - x_2) \\&= (-x_1, -x_2) \in R^2 \\y &= -x\end{aligned}$$

Hence inverse exists.

$\therefore R^2$ is a group under addition.

Commutative Property: Let $x, y \in \mathbb{R}^2$. Then

$$\begin{aligned}x + y &= (x_1, x_2) + (y_1, y_2) \\&= (x_1 + y_1, x_2 + y_2) \\&= (y_1 + x_1, y_2 + x_2) \\&= (y_1, y_2) + (x_1, x_2) \\&= y + x \in \mathbb{R}^2\end{aligned}$$

Hence commutative property is satisfied.

$\therefore \mathbb{R}^2$ is a abelian group under addition.

To prove: $a(x + y) = (ax) + (ay)$

Let $a \in R$ and $x, y \in R^2$

$$\begin{aligned}a(x + y) &= a(x_1 + y_1, x_2 + y_2) \\&= (a(x_1 + y_1), a(x_2 + y_2)) \\&= (ax_1 + ay_1, ax_2 + ay_2) \\&= (ax_1, ax_2) + (ay_1, ay_2) \\&= a(x_1, x_2) + a(y_1, y_2) \\&= ax + ay\end{aligned}$$

Hence it is satisfied.

To prove: $(a + b)(x) = ax + bx$

Let $a, b \in R$ and $x \in R^2$

$$\begin{aligned}(a + b)x &= (a + b)(x_1, x_2) \\&= ((a + b)x_1, (a + b)x_2) \\&= (ax_1 + bx_1, ax_2 + bx_2) \\&= (ax_1, ax_2) + (bx_1, bx_2) \\&= a(x_1, x_2) + b(x_1, x_2) \\&= ax + bx\end{aligned}$$

Hence it is satisfied.

To prove: $(ab)x = a(bx)$

Let $a, b \in R, x \in R^2$

$$\begin{aligned}(ab)(x) &= (ab)(x_1, x_2) \\&= ((ab)x_1, (ab)x_2) \\&= (a(bx_1), a(bx_2)) \\&= a(bx_1, bx_2) \\&= a(bx)\end{aligned}$$

Hence it is satisfied.

To prove: $1 \cdot x = x$

Let $1 \in R$ and $x = (x_1, x_2) \in R^2$

$$\begin{aligned}1 \cdot x &= 1 \cdot (x_1, x_2) \\&= (1 \cdot x_1, 1 \cdot x_2) \\&= (x_1, x_2) \\&= x\end{aligned}$$

Hence it is satisfied.

All the conditions are satisfied. $\therefore R^2$ is a vector space over the field of 

Exercise

- ① Check $V = \{(a_1, a_2, \dots, a_n) \in R^n : a_1, a_2, \dots, a_n \in R\}$ is a vector space over R or not, under two operations define

vector addition: $x + y = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n) \forall$

$x = (a_1, a_2, \dots, a_n), y = (b_1, b_2, \dots, b_n) \in R^n$

scalar multiplication:

$cx = (ca_1, ca_2, \dots, ca_n) \forall x = (a_1, a_2, \dots, a_n) \in R^n \text{ and } c \in R$

- ② Check $V = \{(a_1, a_2, \dots, a_n) \in C^n : a_1, a_2, \dots, a_n \in C\}$ is a vector space over C or not, under two operations define

vector addition: $x + y = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n) \forall$

$x = (a_1, a_2, \dots, a_n), y = (b_1, b_2, \dots, b_n) \in C^n$

scalar multiplication:

$cx = (ca_1, ca_2, \dots, ca_n) \forall x = (a_1, a_2, \dots, a_n) \in C^n \text{ and } c \in C$

Example 2

Check $V = \{(x_1, x_2) \in R^2 : x_1, x_2 \in R\}$ is a vector space over R or not, under two operations define

vector addition:

$$x + y = (x_1 + 2y_1, x_2 + 3y_2) \quad \forall x = (x_1, x_2), y = (y_1, y_2) \in R^2$$

scalar multiplication: $ax = (ax_1, ax_2) \quad \forall x = (x_1, x_2) \in R^2 \text{ and } a \in R$

Solution:

Given that $V = (R^2, +, \cdot)$ and $F = R$.

Let $x = (x_1, x_2), y = (y_1, y_2)$ and $z = (z_1, z_2)$, where

$x_1, x_2, y_1, y_2, z_1, z_2 \in R$.

To prove: V is an abelian group

Closure Axiom: Let $x, y \in R^2$. Then

$$\begin{aligned} x + y &= (x_1, x_2) + (y_1, y_2) \\ &= (x_1 + 2y_1, x_2 + 3y_2) \in R^2 \end{aligned}$$

Thus $x + y \in R^2$. Hence closure property is satisfied.

Associative Property: Let $x, y, z \in R^2$. Then

$$\begin{aligned}x + (y + z) &= (x_1, x_2) + ((y_1, y_2) + (z_1, z_2)) \\&= (x_1, x_2) + (y_1 + 2z_1, y_2 + 3z_2) \\&= (x_1 + 2(y_1 + 2z_1), x_2 + 3(y_2 + 3z_2)) \\&= (x_1 + (2y_1 + 4z_1), x_2 + (3y_2 + 9z_2)) \\&= ((x_1 + 2y_1) + 4z_1, (x_2 + 3y_2) + 9z_2) \\&= ((x_1 + 2y_1), (x_2 + 3y_2)) + (4z_1, 9z_2) \\&\neq (x + y) + z\end{aligned}$$

Hence associative property is not satisfied.

$\therefore R^2$ is not a vector space over the field of real numbers.

Exercise

- ① Check $V = \{(a_1, a_2) \in R^2 : a_1, a_2 \in R\}$ is a vector space over R or not, under two operations define
vector addition: $x + y = (a_1 + b_1, a_2 b_2) \forall x, y \in R^2$
scalar multiplication: $cx = (ca_1, ca_2) \forall x \in R^2$ and $c \in R$
- ② Check $V = \{(a_1, a_2) \in R^2 : a_1, a_2 \in R\}$ is a vector space over R or not, under two operations define
vector addition: $x + y = (a_1 + b_1, a_2 - b_2) \forall x, y \in R^2$
scalar multiplication: $cx = (ca_1, a_2) \forall x \in R^2$ and $c \in R$
- ③ Check $V = \{(a_1, a_2) : a_1, a_2 \in F\}$ is a vector space over F or not, under two operations define
vector addition: $x + y = (a_1 + b_1, a_2 + b_2) \forall x, y \in V$
scalar multiplication: $cx = (a_1, 0) \forall x \in V$ and $c \in F$

Polynomial

- A polynomial with coefficients from a field F is $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_{n-1}x^{n-1} + a_nx^n$ where n is a nonnegative integer and each a_i , called the coefficient of x_i is in F
- Two polynomials, $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_{n-1}x^{n-1} + a_nx^n$ and $g(x) = b_0 + b_1x + b_2x^2 + \dots + b_{m-1}x^{m-1} + b_mx^m$ are called *equal* if $m = n$ and $a_i = b_i$ for $i = 0, 1, \dots, n$
- $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_{n-1}x^{n-1} + a_nx^n$ and $g(x) = b_0 + b_1x + b_2x^2 + \dots + b_{m-1}x^{m-1} + b_mx^m$.
Suppose $m \leq n$ and define $b_{m+1} = b_{m+2} = \dots = b_n = 0$.
Then $g(x)$ can be written as
 $g(x) = b_0 + b_1x + b_2x^2 + \dots + b_{m-1}x^{m-1} + b_mx^m + \dots + b_nx^n$.

Example 3

Check $V = \{ \text{The set of all polynomials of degree } \leq 2 \}$ is a vector space over R or not, under two operations define

Addition of polynomials:

$$f(x) + g(x) = (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 \quad \forall f(x), g(x) \in V$$

Scalar multiplication:

$$\alpha f(x) = \alpha a_0 + \alpha a_1 x + \alpha a_2 x^2 \quad \forall f(x) \in V \text{ and } \alpha \in R$$

Solution:

Given that $V = \{ \text{The set of all polynomials of degree } \leq 2 \}$.

Let $f(x), g(x), h(x) \in V$. i.e.,

$$f(x) = a_0 + a_1 x + a_2 x^2$$

$$g(x) = b_0 + b_1 x + b_2 x^2$$

$$h(x) = c_0 + c_1 x + c_2 x^2$$

where $a_0, a_1, a_2, b_0, b_1, b_2, c_0, c_1, c_2 \in R$ To prove: V is a abelian group

Closure Axiom:

$$\begin{aligned}f(x) + g(x) &= a_0 + a_1x + a_2x^2 + b_0 + b_1x + b_2x^2 \\&= (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 \in V\end{aligned}$$

Hence closure property is satisfied.

Associative Property: $f(x) + [g(x) + h(x)] = [f(x) + g(x)] + h(x)$

$$\begin{aligned}&f(x) + [g(x) + h(x)] \\&= a_0 + a_1x + a_2x^2 + ((b_0 + c_0) + (b_1 + c_1)x + (b_2 + c_2)x^2) \\&= (a_0 + b_0 + c_0) + (a_1 + b_1 + c_1)x + (a_2 + b_2 + c_2)x^2 \\&= ((a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2) + c_0 + c_1x + c_2x^2 \\&= [f(x) + g(x)] + h(x)\end{aligned}$$

Hence associative property is satisfied.

Existence of Identity: Let $f(x) + e = e + f(x) = f(x)$. Then

$$\begin{aligned}f(x) + e &= f(x) \\(a_0 + a_1x + a_2x^2) + (e_0 + e_1x + e_2x^2) &= a_0 + a_1x + a_2x^2 \\e_0 + e_1x + e_2x^2 &= 0 + 0x + 0x^2 \in V\end{aligned}$$

Hence identity exists.

Existence of Inverse: Let $f^{-1}(x)$ is the inverse of $f(x)$. Then

$$\begin{aligned}f(x) + f^{-1}(x) &= e \\a_0 + a_1x + a_2x^2 + f^{-1}(x) &= 0 + 0x + 0x^2 \\f^{-1}(x) &= -(a_0 + a_1x + a_2x^2)\end{aligned}$$

Hence inverse exists. $(V, +)$ is a group.

Commutative Property: $f(x) + g(x) = g(x) + f(x)$

$$\begin{aligned}f(x) + g(x) &= a_0 + a_1x + a_2x^2 + b_0 + b_1x + b_2x^2 \\&= (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 \\&= (b_0 + a_0) + (b_1 + a_1)x + (b_2 + a_2)x^2 \\&= g(x) + f(x)\end{aligned}$$

Hence commutative property is satisfied.

$\therefore (V, +)$ is an abelian group.

To prove: $\alpha(\beta f(x)) = \alpha\beta(f(x))$

Let $\alpha, \beta \in R$ and $f(x) \in V$. Then

$$\begin{aligned}\alpha(\beta f(x)) &= \alpha(\beta a_0 + \beta a_1x + \beta a_2x^2) \\&= \alpha\beta a_0 + \alpha\beta a_1x + \alpha\beta a_2x^2 \\&= \alpha\beta(a_0 + a_1x + a_2x^2) \\&= \alpha\beta(f(x))\end{aligned}$$

Hence it is satisfied.

To prove: $\alpha(f(x) + g(x)) = \alpha f(x) + \alpha g(x)$

Let $\alpha \in R$ and $f(x), g(x) \in V$. Then

$$\begin{aligned}\alpha(f(x) + g(x)) &= \alpha((a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2) \\&= \alpha(a_0 + b_0) + \alpha(a_1 + b_1)x + \alpha(a_2 + b_2)x^2 \\&= \alpha a_0 + \alpha a_1 x + \alpha a_2 x^2 + \alpha b_0 + \alpha b_1 x + \alpha b_2 x^2 \\&= \alpha(a_0 + a_1 x + a_2 x^2) + \alpha(b_0 + b_1 x + b_2 x^2) \\&= \alpha f(x) + \alpha g(x)\end{aligned}$$

Hence it is satisfied.

To prove: $(\alpha + \beta)f(x) = \alpha f(x) + \beta f(x)$

Let $\alpha, \beta \in R$ and $f(x) \in V$. Then

$$\begin{aligned}(\alpha + \beta)f(x) &= (\alpha + \beta)(a_0 + a_1x + a_2x^2) \\&= (\alpha + \beta)a_0 + (\alpha + \beta)a_1x + (\alpha + \beta)a_2x^2 \\&= \alpha a_0 + \beta a_0 + \alpha a_1x + \beta a_1x + \alpha a_2x^2 + \beta a_2x^2 \\&= \alpha(a_0 + a_1x + a_2x^2) + \beta(a_0 + a_1x + a_2x^2) \\&= \alpha f(x) + \beta f(x)\end{aligned}$$

To prove: $1f(x) = f(x)$

Let $1 \in R$ and $f(x) \in V$. Then

$$\begin{aligned}1f(x) &= 1(a_0 + a_1x + a_2x^2) \\&= a_0 + a_1x + a_2x^2 \\&= f(x)\end{aligned}$$

All the conditions are satisfied.

Hence $(V, +, \cdot)$ is a vector space.

Exercise

- ① Check $V = \{ \text{The set of all polynomials of degree } \leq n \}$ is a vector space over R or not, under two operations define

Addition of polynomials:

$$f(x) + g(x) =$$

$$(a_n + b_n)x_n + (a_{n-1} + b_{n-1})x_{n-1} + \dots + (a_0 + b_0) \quad \forall f(x), g(x) \in V$$

Scalar multiplication:

$$cf(x) = ca_nx_n + ca_{n-1}x_{n-1} + \dots + ca_1x_1 + ca_0 \quad \forall f(x) \in V \text{ and } c \in R$$

- ② Check $V = \{ \text{The set of all polynomials} \}$ is a vector space over F or not, under two operations define

Addition of polynomials: $f(x) + g(x) \in V$ is the usual operation of addition of polynomials

Scalar multiplication: $cf(x) \in V$ the usual operation of the product of a scalar c and a polynomial $f(x)$

Example 4

Check $V = M_{2 \times 2}(R)$, is a collection of 2×2 matrices whose entries from R is a vector space or not, under usual addition and multiplication define

Matrix addition:

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix} \quad \forall A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \in V$$

Scalar multiplication:

$$\alpha \cdot A = \begin{bmatrix} \alpha a_{11} & \alpha a_{12} \\ \alpha a_{21} & \alpha a_{22} \end{bmatrix} \quad \forall \alpha \in R \text{ and } A \in V$$

Solution:

Given that $V = M_{2 \times 2}(R)$ is a collection of 2×2 of matrix.

To prove: V is a abelian group

Closure Axiom: Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$,

$$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \in V$$

$$\begin{aligned} A + B &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \\ &= \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix} \in V \end{aligned}$$

Hence closure property is satisfied.

Associative Property: $A + (B + C) = (A + B) + C$

$$B + C = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} + \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} = \begin{bmatrix} b_{11} + c_{11} & b_{12} + c_{12} \\ b_{21} + c_{21} & b_{22} + c_{22} \end{bmatrix}$$

$$\begin{aligned}
 A + (B + C) &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} b_{11} + c_{11} & b_{12} + c_{12} \\ b_{21} + c_{21} & b_{22} + c_{22} \end{bmatrix} \\
 &= \begin{bmatrix} a_{11} + b_{11} + c_{11} & a_{12} + b_{12} + c_{12} \\ a_{21} + b_{21} + c_{21} & a_{22} + b_{22} + c_{22} \end{bmatrix}
 \end{aligned} \tag{1}$$

$$\begin{aligned}
 A + B &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix} \\
 (A + B) + C &= \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix} + \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \\
 &= \begin{bmatrix} a_{11} + b_{11} + c_{11} & a_{12} + b_{12} + c_{12} \\ a_{21} + b_{21} + c_{21} & a_{22} + b_{22} + c_{22} \end{bmatrix}
 \end{aligned} \tag{2}$$

From equations (1) and (2), we have

$$A + (B + C) = (A + B) + C$$

Hence associative property is satisfied.

Existence of Identity:

$$A + E = E + A = A$$

Let $E = \begin{bmatrix} e_1 & e_2 \\ e_3 & e_4 \end{bmatrix}$ be identity element.

$$\begin{aligned} A + E &= A \\ \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} e_1 & e_2 \\ e_3 & e_4 \end{bmatrix} &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\ \begin{bmatrix} e_1 & e_2 \\ e_3 & e_4 \end{bmatrix} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

Hence identity exists.

Existence of Inverse: $A + A^{-1} = A^{-1} + A = E$

Let $A, E \in V$

$$\begin{aligned} A + A^{-1} &= E \\ \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + A^{-1} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\ A^{-1} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\ &= - \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \end{aligned}$$

Hence inverse exists.

$(V, +,)$ is a group.

Commutative Property: $A + B = B + A$

$$\begin{aligned} A + B &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \\ &= \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix} \\ &= \begin{bmatrix} b_{11} + a_{11} & b_{12} + a_{12} \\ b_{21} + a_{21} & b_{22} + a_{22} \end{bmatrix} \\ &= \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} + \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\ &= B + A \end{aligned}$$

Hence commutative property is satisfied.

$(V, +)$ is an abelian group.

To prove: $\alpha(A + B) = \alpha A + \alpha B$

Let $\alpha, \in R$ and $A, B \in V$. Then

$$\begin{aligned}\alpha(A + B) &= \alpha \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix} \\ &= \begin{bmatrix} \alpha(a_{11} + b_{11}) & \alpha(a_{12} + b_{12}) \\ \alpha(a_{21} + b_{21}) & \alpha(a_{22} + b_{22}) \end{bmatrix} \\ &= \begin{bmatrix} \alpha a_{11} + \alpha b_{11} & \alpha a_{12} + \alpha b_{12} \\ \alpha a_{21} + \alpha b_{21} & \alpha a_{22} + \alpha b_{22} \end{bmatrix} \\ &= \alpha \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \alpha \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \\ &= \alpha A + \alpha B\end{aligned}$$

To prove: $(\alpha + \beta)A = \alpha A + \beta A$

Let $\alpha, \beta \in R$ and $A, B \in V$. Then

$$\begin{aligned}(\alpha + \beta)A &= (\alpha + \beta) \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\&= \begin{bmatrix} (\alpha + \beta)a_{11} & (\alpha + \beta)a_{12} \\ (\alpha + \beta)a_{21} & (\alpha + \beta)a_{22} \end{bmatrix} \\&= \begin{bmatrix} \alpha a_{11} + \beta a_{11} & \alpha a_{12} + \beta a_{12} \\ \alpha a_{21} + \beta a_{21} & \alpha a_{22} + \beta a_{22} \end{bmatrix} \\&= \alpha \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + \beta \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\&= \alpha A + \beta A\end{aligned}$$

To prove: $\alpha(\beta A) = (\alpha\beta)A$

Let $\alpha, \beta \in R$ and $A \in V$. Then

$$\begin{aligned}\alpha(\beta A) &= \alpha \begin{bmatrix} \beta a_{11} & \beta a_{12} \\ \beta a_{21} & \beta a_{22} \end{bmatrix} = \begin{bmatrix} \alpha\beta a_{11} & \alpha\beta a_{12} \\ \alpha\beta a_{21} & \alpha\beta a_{22} \end{bmatrix} \\ &= (\alpha\beta) \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \in V\end{aligned}$$

To prove: $1A = A$

Let $1 \in R$ and $A \in V$. Then

$$\begin{aligned}1A &= 1 \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\ &= A\end{aligned}$$

All the conditions are satisfied.

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Exercise

Check $V = \{M_{m \times n}(F)\}$, is the set of all $m \times n$ matrices whose entries from F is a vector space or not, with the following operations

Matrix addition: $(A + B)_{ij} = A_{ij} + B_{ij} \quad \forall A, B \in M_{m \times n}(F)$

Scalar multiplication: $(cA)_{ij} = cA_{ij} \quad \forall A \in M_{m \times n}(F) \text{ and } c \in F$
for $1 \leq i \leq m$ and $1 \leq j \leq n$.

Cancellation Law for Vector Addition

If x, y, z are vectors in a vector space V such that $x + z = y + z$, then $x = y$.

Proof:

Given: $x, y, z \in V$ and $x + z = y + z$.

To prove: $x = y$.

Now

$$\begin{aligned}x &= x + 0 && (\because \text{Identity}) \\&= x + (z + v) && (\because \text{Inverse}) \\&= (x + z) + v && (\because \text{Associativity}) \\&= (y + z) + v && (\because \text{Given}) \\&= y + (z + v) && (\because \text{Associativity}) \\&= y + 0 && (\because \text{Inverse}) \\&= y\end{aligned}$$

Hence proved.

Corollary 1

The additive identity vector 0 described in (VS 3) is unique.

Corollary 2

The additive inverse vector y described in (VS 4) is unique.

Theorem

In any vector space V , the following statements are true:

- i. $0x = 0 \quad \forall x \in V$
- ii. $(-a)x = -(ax) = a(-x) \quad \forall a \in F \text{ and } x \in V$
- iii. $a0 = 0 \quad \forall a \in F$

Proof:

- i. Let $x \in V$ and $0 \in F$, then $0x \in V$.

Now,

$$0x = 0x + 0 \quad (3)$$

$$\begin{aligned} \text{Also, } 0x &= (0 + 0)x \\ &= 0x + 0x \end{aligned} \quad (4)$$

From (3) and (4)

$$0x + 0 = 0x + 0x$$

$$0x = 0$$

ii. Let $x \in V$ then $ax \in V$.

Now,

$$\begin{aligned} ax + (-a)x &= [a + (-a)]x & (5) \\ &= 0x \\ &= 0 \quad (\text{by (i)}) \end{aligned}$$

$$(-a)x = -(ax) \quad (6)$$

$$\text{In particular, } (-1)x = -x \quad (7)$$

$$\begin{aligned} a(-x) &= a[-1 \cdot x] \\ &= (-a)x & (8) \end{aligned}$$

From (6) and (8)

$$(-a)x = -(ax) = a(-x)$$

iii. Let $a \in F$ and $0 \in V$.
Then

$$\begin{aligned}a0 + a0 &= a(0 + 0) \\&= a0 \\&= 0 + a0 \quad (\because \text{identity}) \\a0 &= 0 \quad (\because \text{cancellation law})\end{aligned}$$

Hence proved.

SUBSPACES

Subspace

A non-empty subset W of vector space V over a field F is called a **subspace** of V , if W is a vector space over F with the same operations defined in V .

Note

Every vector space V has at least two subspaces.

- 1 V itself is a subspace of V
- 2 $\{0\}$ is a subspace of V , is called the **zero subspace** of V .

These two subspaces are called **trivial subspaces**.

Example

Let $V = R^3 = \{(a, b, c) : a, b, c \in R\}$ be a vector space over R with usual addition and scalar multiplication. Then $W = \{(a, b, 0) : a, b \in R\}$ is a subspace.

Note

W is a subspace of V if and only if the following properties holds:

- i. $x + y \in W$, for each pair of elements $x, y \in W$
- ii. $cx \in W$, whenever $c \in F$ and $x \in W$
- iii. There exists an elements $0 \in W$ such that $0 + x = x + 0 = x$, for each $x \in W$
- iv. For each $x \in W$, there exists an element $-x \in W$ such that

$$x + (-x) = 0 = (-x) + x$$

Theorem

Let V be a vector space and W a subset of V . Then W is a subspace of V if and only if the following three conditions hold for the operations defined in V .

- i. $0 \in W$
- ii. $x + y \in W$, whenever $x \in W$ and $y \in W$
- iii. $cx \in W$, whenever $c \in F$ and $x \in W$

Proof:

Assume W is a subspace of V . Then we have to prove [i], [ii], [iii] holds.

Since W is a subspace, we have W itself is a vector space with the operations of addition and scalar multiplication defined on V .

Let $x, y \in W$ and $c \in F$, then

$$x + y \in W \text{ and } cx \in W$$

Hence [ii], [iii] holds.

Let $0' \in W$ be the zero element of W .

Then for any $x \in W$,

$$x + 0' = 0' + x = x \quad \forall x \in W \quad (1)$$

Also, $x \in W \Rightarrow x \in V$

$$\Rightarrow x + 0 = 0 + x = x \quad (2)$$

From (1) and (2)

$$x + 0' = x + 0$$

$$0' = 0 \quad (\text{by cancellation law})$$

Hence, $0 \in W$

Conversely,

Assume the conditions [i], [ii], [iii] hold, then we have to prove W is a subspace.

So, it is enough to prove there exists the additive inverse of each vector in W .

But, if $x \in W$, then $(-1)x \in W$, by condition (iii)

We know that by Theorem

"Any vector space V , we have $(-a)x = -(ax) = a(-x)$ for each $a \in F$ and $x \in V$ "

Using above fact, we have

$$(-1)x = -x \in W$$

$\therefore$ Additive inverse exists

Hence W is a subspace of V .

Theorem

Any nonempty subset W of a vector space V over F is a subspace of V if and only if $\alpha w_1 + \beta w_2 \in W \quad \forall \quad w_1, w_2 \in W \quad \text{and} \quad \alpha, \beta \in F$.

Example 1

Prove that $W = \{(x, 0) \in R^2 : x \in R\}$ is a subspace of the vector space $V = R^2$ under the operation of addition and scalar multiplication.

Solution: Given that $V = R^2$ vector space over R and

$$W = \{(x, 0), x \in R\}$$

To prove: W is a subspace of V .

i.e., To check $\alpha w_1 + \beta w_2 \in W$.

Clearly $W \subset V$.

Let $w_1 = (x_1, 0), w_2 = (x_2, 0) \in W$ and $\alpha, \beta \in R$.

Then

$$\begin{aligned}\alpha w_1 + \beta w_2 &= \alpha(x_1, 0) + \beta(x_2, 0) \\ &= (\alpha x_1, 0) + (\beta x_2, 0) \\ &= (\alpha x_1 + \beta x_2, 0) \in W\end{aligned}$$

Hence W is a subspace of V .

Example 2

Prove that $W = \{(x, y, 0) \in R^3 : x, y \in R\}$ is a subspace of $V = R^3$ under the operation of addition and scalar multiplication.

Solution: Given $V = R^3$ and $W = \{(x, y, 0) \in R^3 : x, y \in R\}$.

It is clear that $0 = (0, 0, 0) \in W$.

Let $x, y \in W$

$$x = (x_1, x_2, 0)$$

$$y = (y_1, y_2, 0)$$

$$\begin{aligned}x + y &= (x_1, x_2, 0) + (y_1, y_2, 0) \\&= (x_1 + y_1, x_2 + y_2, 0) \in W\end{aligned}$$

$$\begin{aligned}\alpha x &= \alpha(x_1, x_2, 0) \\&= (\alpha x_1, \alpha x_2, 0) \in W\end{aligned}$$

Hence W is a subspace of V .

Example 3

Let $V = R^3$ is a vector space. Let $w = \{(x_1, x_2, x_3) \in R^3 : x_1 = x_3 + 2\}$. Determine whether W is subspace of V under the operation of addition and scalar multiplication.

Solution: Let $x, y \in W$

$$x = (x_1, x_2, x_3) \Rightarrow x_1 - x_3 - 2 = 0$$

$$y = (y_1, y_2, y_3) \Rightarrow y_1 - y_3 - 2 = 0$$

$$\begin{aligned}x + y &= (x_1, x_2, x_3) + (y_1, y_2, y_3) \\&= (x_1 + y_1, x_2 + y_2, x_3 + y_3) \\&= x_1 + y_1 - (x_3 + y_3) - 2 \\&= (x_1 - x_3) + (y_1 - y_3) - 2 \\&= (x_1 - x_3 - 2) + (y_1 - y_3) = 0 + 2 \\&= 2 \neq 0\end{aligned}$$

$$x + y \notin W$$

Hence W is not a subspace of V .

Example 4

Determine whether the W be the set of all 2×3 matrices with the second row zero is a subspace of a vector space $V = M_{2 \times 3}(R)$.

Solution: Given that $V = M_{2 \times 3}(R)$ and

$$W = \left\{ \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & 0 & 0 \end{pmatrix} \in M_{2 \times 3}(R) : a_{11}, a_{12}, a_{13} \in R \right\}$$

Let $w_1 = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & 0 & 0 \end{pmatrix}$, $w_2 = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ 0 & 0 & 0 \end{pmatrix} \in W$. Then

$$\begin{aligned} \alpha w_1 + \beta w_2 &= \alpha \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & 0 & 0 \end{pmatrix} + \beta \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ 0 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} \alpha a_{11} + \beta b_{11} & \alpha a_{12} + \beta b_{12} & \alpha a_{13} + \beta b_{13} \\ 0 & 0 & 0 \end{pmatrix} \in W \end{aligned}$$

Hence W is a subspace of V .

Example 5

Determine set of all 3×3 diagonal matrix is a subspace of $V = M_{3 \times 3}(R)$ under the operations of matrix addition any scalar multiplication.

Solution: Given $V = M_{3 \times 3}(R)$ and

$$W = \left\{ \begin{bmatrix} \lambda_{11} & 0 & 0 \\ 0 & \lambda_{22} & 0 \\ 0 & 0 & \lambda_{33} \end{bmatrix} \in M_{3 \times 3}(R) \right\}$$

Let $w_1, w_2 \in W$, and $\alpha, \beta \in F$. Then

$$w_1 = \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix} \in M_{3 \times 3}(R)$$
$$w_2 = \begin{bmatrix} b_{11} & 0 & 0 \\ 0 & b_{22} & 0 \\ 0 & 0 & b_{33} \end{bmatrix} \in M_{3 \times 3}(R)$$

$$\begin{aligned}
\alpha w_1 + \beta w_2 &= \alpha \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix} + \beta \begin{bmatrix} b_{11} & 0 & 0 \\ 0 & b_{22} & 0 \\ 0 & 0 & b_{33} \end{bmatrix} \\
&= \begin{bmatrix} \alpha a_{11} & 0 & 0 \\ 0 & \alpha a_{22} & 0 \\ 0 & 0 & \alpha a_{33} \end{bmatrix} + \begin{bmatrix} \beta b_{11} & 0 & 0 \\ 0 & \beta b_{22} & 0 \\ 0 & 0 & \beta b_{33} \end{bmatrix} \\
&= \begin{bmatrix} \alpha a_{11} + \beta b_{11} & 0 & 0 \\ 0 & \alpha a_{22} + \beta b_{22} & 0 \\ 0 & 0 & \alpha a_{33} + \beta b_{33} \end{bmatrix} \in W
\end{aligned}$$

Hence W is subspace of V .

Example 6

Prove that $W = \{(a_1, a_2, a_3) \in R^3 : 2a_1 - 7a_2 + a_3 = 0\}$, is a subspace of is a vector space $V = R^3$ under the operation of addition and scalar multiplication.

Solution: It is clear that $0 \in W$

Let $x, y \in W$

$$x = (x_1, x_2, x_3) \Rightarrow 2x_1 - 7x_2 + x_3 = 0$$

$$y = (y_1, y_2, y_3) \Rightarrow 2y_1 - 7y_2 + y_3 = 0$$

$$\begin{aligned}x + y &= (x_1, x_2, x_3) + (y_1, y_2, y_3) \\&= (x_1 + y_1, x_2 + y_2, x_3 + y_3) \\&= 2(x_1 + y_1) - 7(x_2 + y_2) + (x_3 + y_3) \\&= 0\end{aligned}$$

$$x + y \in W$$

$$\begin{aligned}\alpha x &= \alpha(x_1, x_2, x_3) \\ &= \alpha(2x_1 - 7x_2 + x_3) \\ &= 0 \\ \alpha x &\in W\end{aligned}$$

Hence W is a subspace of V .

Exercise:

- ① Is the set $W = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 = x_2^2, x_1, x_2 \in \mathbb{R}\}$ is a subspace of the vector space $V = \mathbb{R}^2$ under the operations of addition and scalar multiplication?
- ② Determine the set $W = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : a_1 + a_2 = 0\}$ is a subspace of $V = \mathbb{R}^3$ under the operation of addition and scalar multiplication.
- ③ Determine the set of all 3×3 lower triangular matrices is subspace of $V = M_{3 \times 3}(\mathbb{R})$ under the operations of matrix addition and scalar multiplication.
- ④ Let V be the vector space of all square matrices over $\mathbb{R}$. Determine the set $W = \{A : A \in V, A = A^t\}$ is a subspace of V or not.

Theorem

Intersection of two subspaces of a vector space V over a field F is a subspace of V .

Proof:

Given V is a vector space over F .

Let W_1 and W_2 be two subspaces of V .

Then we have to prove that $W_1 \cap W_2$ is a subspace of V .

Since W_1 and W_2 are subspaces, we have

$$0 \in W_1 \quad \text{and} \quad 0 \in W_2$$

$$\therefore 0 \in W_1 \cap W_2$$

Hence, $W_1 \cap W_2$ is non empty.

Let $x, y \in W_1 \cap W_2$.

Then $x, y \in W_1$ and $x, y \in W_2$.

Since W_1 and W_2 are subspaces, we have

$$x + y \in W_1 \text{ and } x + y \in W_2 \Rightarrow x + y \in W_1 \cap W_2$$

Let $x \in W_1 \cap W_2$ and $c \in F$.

Then $x \in W_1$ and $x \in W_2$.

Since W_1 and W_2 are subspaces, we have

$$cx \in W_1 \text{ and } cx \in W_2 \Rightarrow cx \in W_1 \cap W_2$$

Hence $W_1 \cap W_2$ is a subspace of V .

Remark

Intersection of a collection of subspaces of V over F is a subspace of V .

Theorem

Let W_1 and W_2 be subspaces of $V(F)$. Then $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Proof: Given W_1 and W_2 be subspaces of V .

Let $W_1 \cup W_2$ is a subspace of V .

To prove: $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Let $W_1 \not\subseteq W_2$ and $W_2 \not\subseteq W_1$.

$$W_1 \not\subseteq W_2 \Rightarrow \text{there exists an element } u \in W_1 \text{ and } u \notin W_2 \quad (3)$$

$$W_2 \not\subseteq W_1 \Rightarrow \text{there exists an element } v \in W_2 \text{ and } v \notin W_1 \quad (4)$$

From (3) and (4), we have

$$u \in W_1 \cup W_2 \quad \text{and} \quad v \in W_1 \cup W_2.$$

Since $W_1 \cup W_2$ is a subspace of V , we have $u + v \in W_1 \cup W_2$.

$$\therefore u + v \in W_1 \quad \text{or} \quad u + v \in W_2.$$

If $u + v \in W_1$ and $u \in W_1$, then $u + v - u \in W_1 \Rightarrow v \in W_1$, which is the contradiction.

If $u + v \in W_2$ and $v \in W_2$, then $u + v - v \in W_2 \Rightarrow u \in W_2$, which is the contradiction.

Hence our assumption $W_1 \not\subseteq W_2$ and $W_2 \not\subseteq W_1$ is wrong.
 $\therefore W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Conversely,

Let $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

To prove: $W_1 \cup W_2$ is a subspace.

If $W_1 \subseteq W_2$, then $W_1 \cup W_2 = W_2$, which is a subspace.

If $W_2 \subseteq W_1$, then $W_1 \cup W_2 = W_1$, which is a subspace.

$\therefore W_1 \cup W_2$ is a subspace.

Hence proved.

LINEAR COMBINATIONS

Linear Combination

Let V be a vector space and S be a non empty subset of V . A vector $v \in V$ is called a **linear combination** of vectors of S if there exists a finite number of vectors $u_1, u_2, u_3, \dots, u_n \in S$ and scalars $a_1, a_2, a_3, \dots, a_n \in F$ such that

$$v = a_1u_1 + a_2u_2 + a_3u_3 + \dots + a_nu_n$$

Note

- In any vector space V , $0 \cdot v = 0 \quad \forall \quad v \in V$.
i.e., Zero vector is a linear combination of any non empty subset of V .
- Consider $(2, 1)$, which is the linear combination of $(1, 0)$ & $(0, 1)$.
i.e., $(2, 1) = 2(1, 0) + 1(0, 1)$

Example 1

Let $V = \mathbb{R}^2$; $S = \{(1, 2), (2, 1)\}$. Then, check $v = (2, 2) \in V$ a linear combination of S or not?

Solution:

Let $a, b \in \mathbb{R}$. Then

$$(2, 2) = a(1, 2) + b(2, 1)$$

$$= (a, 2a) + (2b, b)$$

$$= (a + 2b, 2a + b)$$

$$a + 2b = 2 \quad (1)$$

$$2a + b = 2 \quad (2)$$

Solve (1) and (2), we get $a = \frac{2}{3}, b = \frac{2}{3}$.

$$\therefore (2, 2) = \frac{2}{3}(1, 2) + \frac{2}{3}(2, 1).$$

Hence linear combination exist.

Example 2

Check whether the vector $v = (2, -5, 3)$ is a linear combination of the vectors $e_1 = (1, -3, 2)$, $e_2 = (2, -4, -1)$ and $e_3 = (1, -5, 7)$ or not?

Solution: Let $a, b, c \in \mathbb{R}$. Then

$$\begin{aligned}(2, -5, 3) &= ae_1 + be_2 + ce_3 \\&= a(1, -3, 2) + b(2, -4, -1) + c(1, -5, 7) \\&= (a, -3a, 2a) + (2b, -4b, -b) + (c, -5c, 7c) \\&= (a + 2b + c, -3a - 4b - 5c, 2a - b + 7c)\end{aligned}$$

$$a + 2b + c = 2 \tag{3}$$

$$-3a - 4b - 5c = -5 \tag{4}$$

$$2a - b + 7c = 3 \tag{5}$$

The above equations are written as follows

$$\begin{bmatrix} 1 & 2 & 1 \\ -3 & -4 & -5 \\ 2 & -1 & 7 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 2 \\ -5 \\ 3 \end{bmatrix}$$

Solve above system of equations by Cramer's rule.

Let

$$\begin{aligned}\Delta &= \begin{vmatrix} 1 & 2 & 1 \\ -3 & -4 & -5 \\ 2 & -1 & 7 \end{vmatrix} \\ &= 1(-28 - 5) - 2(-21 + 10) + 1(3 + 8) \\ &= -33 + 22 + 11 \\ &= 0\end{aligned}$$

$\Rightarrow \Delta$ is inconsistent.

$\therefore$ There is no solution.

Hence linear combination does not exist.

Example 3

For which values of k will the vector $v = (1, -2, k)$ in R^3 be a linear combination of $u = (3, 0, -2)$ and $w = (2, -1, 5)$.

Solution: Let $a, b \in R$. Then

$$\begin{aligned}(1, -2, k) &= au + bw \\ &= a(3, 0, -2) + b(2, -1, 5) \\ &= (3a, 0, -2a) + (2b, -b, 5b) \\ &= (3a + 2b, -b, -2a + 5b) \\ 3a + 2b &= 1 \tag{5}\end{aligned}$$

$$-b = -2 \tag{6}$$

$$-2a + 5b = k \tag{7}$$

From (5) and (6), we get $a = -1$, $b = 2$.

$\therefore a, b$ exists.

From (7), $k = 12$.

Example 4

Write down the matrix $E = \begin{pmatrix} 3 & 1 \\ 1 & -1 \end{pmatrix}$ as a linear combination of the matrices $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$ and $C = \begin{pmatrix} 0 & 2 \\ 0 & -1 \end{pmatrix}$

Solution: Let $a, b, c \in \mathbb{R}$.

Then

$$\begin{aligned} E &= aA + bB + cC \\ \begin{pmatrix} 3 & 1 \\ 1 & -1 \end{pmatrix} &= a \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} + c \begin{pmatrix} 0 & 2 \\ 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} a & a \\ a & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ b & b \end{pmatrix} + \begin{pmatrix} 0 & 2c \\ 0 & -c \end{pmatrix} \\ &= \begin{pmatrix} a & a + 2c \\ a + b & b - c \end{pmatrix} \end{aligned}$$

$$\begin{aligned}a &= 3 \\a + 2c &= 1 \\a + b &= 1 \\b - c &= -1\end{aligned}$$

Solve above equations, we get

$$\begin{aligned}a &= 3 \\b &= -2 \\c &= -1\end{aligned}$$

$\therefore a, b, c$ exists.

Hence the linear combination of E is $3A - 2B - C$

Span

Let S be a nonempty subset of a vector space V . The **span** of S , denoted **$\text{span}(S)$** , is the set consisting of all linear combinations of the vectors in S .

Note

For convenience, we define $\text{span}(\phi) = \{0\}$.

Generate

A subset S of a vector space V **generates** or **spans** V , if **$\text{span}(S) = V$** .

We also say that the vectors of S *generate or span* V .

Example 1

Determine whether the vector $v = (x, 1)$ belong to $\text{span}(S)$, where $S = \{(1, 1), (1, 0)\}$.

Solution:

Let $a, b \in F$. Then

$$a(1, 1) + b(1, 0) = (x, 1)$$

$$(a, a) + (b, 0) = (x, 1)$$

$$(a + b, a) = (x, 1)$$

$$\Rightarrow a + b = x$$

$$a = 1$$

Solving the above equations, we get $a = 1, \quad b = x - 1$

$$\therefore v = 1(1, 1) + (x - 1)(1, 0)$$

Hence v is spanned by S .

Example 2

Determine whether the vector $v = 2x^3 - x^2 + x + 3$ belong to $\text{span}(S)$, where $S = \{x^3 + x^2 + x + 1, x^2 + x + 1, x + 1\}$.

Solution: Let $a, b, c \in F$. Then

$$\begin{aligned}a(x^3 + x^2 + x + 1) + b(x^2 + x + 1) + c(x + 1) &= v \\(ax^3 + ax^2 + ax + a) + (bx^2 + bx + b) + (cx + c) &= 2x^3 - x^2 + x + 3 \\ax^3 + (a + b)x^2 + (a + b + c)x + (a + b + c) &= 2x^3 - x^2 + x + 3\end{aligned}$$

Comparing the coefficients of $x^3, x^2, x, \text{constant}$, we get

$$a = 2 \tag{8}$$

$$a + b = -1 \tag{9}$$

$$a + b + c = 1 \tag{10}$$

$$a + b + c = 3 \tag{11}$$

Solving above equations, we get $a = 2, b = -3, c = 2$

Substitute in (11), we get $a + b + c = 2 - 3 + 2 = 1 \neq 3$

$\therefore v$ is not a linear combination of S . Hence S is doesn't span v .

Example 3

Show that the vectors $(1, 1, 0)$, $(1, 0, 1)$ and $(0, 1, 1)$ generate F^3 .

Solution: Let $a, b, c \in F$.

Then

$$a(1, 1, 0) + b(1, 0, 1) + c(0, 1, 1) = (x, y, z)$$

$$(a, a, 0) + (b, 0, b) + (0, c, c) = (x, y, z)$$

$$(a + b, a + c, b + c) = (x, y, z)$$

$$\Rightarrow a + b = x$$

$$a + c = y$$

$$b + c = z$$

Solving the above equations, we get

$$a = \frac{1}{2}(x + y - z)$$

$$b = \frac{1}{2}(x - y + z)$$

$$c = \frac{1}{2}(-x + y + z)$$

$$(x, y, z) = \frac{1}{2}(x + y - z)(1, 1, 0) + \frac{1}{2}(x - y + z)(1, 0, 1) + \frac{1}{2}(-x + y + z)(0, 1, 1)$$

Hence every vector in F^3 is spanned by S .

i.e., $F^3 = \text{Span}(S)$

$\therefore$ The vectors $(1, 1, 0)$, $(1, 0, 1)$ and $(0, 1, 1)$ generate F^3

Example 4

Show that the matrices $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ generate $M_{2 \times 2}(F)$

Solution: Let $a, b, c, d \in F$. Then

$$\begin{pmatrix} a & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} x & y \\ z & w \end{pmatrix}$$

$$\begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ c & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix} = \begin{pmatrix} x & y \\ z & w \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} x & y \\ z & w \end{pmatrix}$$

$$x \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + y \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + z \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + w \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} x & y \\ z & w \end{pmatrix}$$

$$\therefore M_{2 \times 2}(F) = \text{span} \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

$\therefore$ Given vectors generates $M_{2 \times 2}(F)$.

Theorem

The span of any subset S of a vector space V is a subspace of V .
Moreover, any subspace of V that contains S must also contain the span of S .

Proof:

Let $S = \phi$, then $\text{span}(\phi) = \{0\}$.

$\therefore \text{span}(\phi)$ is a subspace of V ($\because \{0\}$ is a subspace that is contained in any subspace of V)

Let $S \neq \phi$, then S contains a vector z .

Now, $0z = 0 \in \text{span}(S)$.

Let $x, y \in \text{span}(S)$.

Then there exist vectors $u_1, u_2, \dots, u_m, v_1, v_2, \dots, v_n \in S$ and scalars $a_1, a_2, \dots, a_m, b_1, b_2, \dots, b_n$ such that

$$x = a_1u_1 + a_2u_2 + \dots + a_mu_m$$

$$y = b_1v_1 + b_2v_2 + \dots + b_nv_n$$

Then

$$\begin{aligned}x + y &= a_1u_1 + a_2u_2 + \cdots + a_mu_m + b_1v_1 + b_2v_2 + \cdots + b_nv_n \\ &\in \text{span}(S)\end{aligned}$$

For any scalar c ,

$$\begin{aligned}cx &= c(a_1u_1 + a_2u_2 + \cdots + a_mu_m) \\ &= (ca_1)u_1 + (ca_2)u_2 + \cdots + (ca_m)u_m \\ &\in \text{span}(S)\end{aligned}$$

$\therefore x + y$ and cx are in $\text{span}(S)$.

Thus $\text{span}(S)$ is a subspace of V .

Let W denote any subspace of V that contains S . i.e., $S \subseteq W$

To prove: $\text{span}(S) \subseteq W$

Let $w \in \text{span}(S)$.

Then $w = c_1w_1 + c_2w_2 + \cdots + c_kw_k$ for some vectors $w_1, w_2, \cdots, w_k \in S$ and some scalars $c_1, c_2, \cdots, c_k$.

Since $S \subseteq W$, we have $w_1, w_2, \cdots, w_k \in W$.

We know that "If W is a subspace of a vector space V and $w_1, w_2, \cdots, w_n \in W$, then $a_1w_1 + a_2w_2 + \cdots + a_nw_n \in W$ for any scalars $a_1, a_2, \cdots, a_n$ ".

By the above fact, we have

$$c_1w_1 + c_2w_2 + \cdots + c_kw_k \in W$$

$$\Rightarrow w \in W$$

$$\therefore \text{span}(S) \subseteq W$$

Hence proved.

LINEAR INDEPENDENT SETS AND LINEAR DEPENDENT SETS

Linearly Dependent Set

A subset S of a vector space $V(F)$ is called **linearly dependent set** if there exists a finite numbers of vectors $v_1, v_2, \dots, v_n \in S$ and scalars $a_1, a_2, \dots, a_n \in F$ such that $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$, for some $a_i \neq 0$.

Also, we say that the vectors of S are linearly dependent.

Linearly Independent Set

A subset S of a vector space $V(F)$ is called **linearly independent set** if there exists a finite numbers of vectors $v_1, v_2, \dots, v_n \in S$ and scalars $a_1, a_2, \dots, a_n \in F$ such that $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$, for all $a_i = 0$.

Also, we say that the vectors of S are linearly independent.

Note

For any vector $v_1, v_2, \dots, v_n$, we have

$$a_1 v_1 + a_2 v_2, \dots, a_n v_n = 0 \quad (1)$$

If $a_1 = a_2 = \dots = a_n = 0$, then (1) is called the *trivial representation* of 0 as a linear combination of $v_1, v_2, \dots, v_n$.

Otherwise *non trivial representation* of 0 as a linear combination of $v_1, v_2, \dots, v_n$.

Example 1

Check whether the set $S = \{v_1, v_2, v_3\}$ is linearly independent or not, where $v_1 = (2, 4, 14)$, $v_2 = (7, -3, 15)$ and $v_3 = (-1, 4, 7)$.

Solution:

Given $v_1 = (2, 4, 14)$, $v_2 = (7, -3, 15)$, $v_3 = (-1, 4, 7)$

Let a, b, c be scalars.

Then

$$av_1 + bv_2 + cv_3 = 0$$

$$a(2, 4, 14) + b(7, -3, 15) + c(-1, 4, 7) = (0, 0, 0)$$

$$(2a + 7b - c, 4a - 3b + 4c, 14a + 15b + 7c) = (0, 0, 0)$$

$$\Rightarrow 2a + 7b - c = 0 \quad (1)$$

$$4a - 3b + 4c = 0 \quad (2)$$

$$14a + 15b + 7c = 0 \quad (3)$$

Method I

Eliminate a from equations (1) and (2), we get

$$17b - 6c = 0 \quad (\text{A})$$

Eliminate a from equations (1) and (3), we get

$$34b - 14c = 0 \Rightarrow 17b - 7c = 0 \quad (\text{B})$$

Solving (A) and (B), we get

$$c = 0 \text{ and } b = 0$$

From (1), we get $a = 0$.

$$\therefore a = 0, b = 0, c = 0$$

Hence $S = \{v_1, v_2, v_3\}$ is linearly independent.

Method II

$$\begin{aligned}\Delta &= \begin{vmatrix} 2 & 7 & -1 \\ 4 & -3 & 4 \\ 14 & 15 & 7 \end{vmatrix} \\ &= 2(-21 - 60) - 7(28 - 56) - 1(60 + 42) \\ &= -162 + 196 - 102 \\ &= -68\end{aligned}$$

$$\therefore \Delta \neq 0$$

$S = \{v_1, v_2, v_3\}$ is linearly independent.

Hence proved.

Example 2

Prove that the set $S = \{v_1, v_2, v_3\}$ is linearly independent, where $v_1 = (0, 1, 1, 0)$, $v_2 = (0, 1, 0, 1)$ and $v_3 = (0, 0, 0, 2)$.

Solution:

Given $v_1 = (0, 1, 1, 0)$, $v_2 = (0, 1, 0, 1)$, $v_3 = (0, 0, 0, 2)$

Let a, b, c be scalars. Then

$$\begin{aligned}av_1 + bv_2 + cv_3 &= 0 \\a(0, 1, 1, 0) + b(0, 1, 0, 1) + c(0, 0, 0, 2) &= (0, 0, 0, 0) \\(a, b, a, b + 2c) &= (0, 0, 0, 0) \\\Rightarrow a &= 0 \\b &= 0 \\b + 2c &= 0 \\\therefore a = 0, b = 0, c &= 0\end{aligned}$$

$S = \{v_1, v_2, v_3\}$ is linearly independent.

Hence proved.

Example 3

Check whether $S = \{1 + x + x^2, 1 + 3x + 2x^2, x\}$ is linearly independent or not.

Solution:

Let a, b, c be scalars. Then

$$\begin{aligned} a \cdot (1 + x + x^2) + b \cdot (1 + 3x + 2x^2) + c \cdot x &= 0 \\ (a + b) \cdot 1 + (a + 3b + c) \cdot x + (a + 2b) \cdot x^2 &= 0 \cdot 1 + 0 \cdot x + 0 \cdot x^2 \end{aligned}$$

Comparing the coefficients, we get

$$\begin{aligned} a + b &= 0 \\ a + 3b + c &= 0 \\ a + 2b &= 0 \end{aligned}$$

Solving the equations, we get $a = 0, b = 0$ and $c = 0$
 S is linearly dependent.

Example 4

Check whether the set $S = \{(1, 0, 0), (0, 1, 0), (1, 1, 1), (1, 1, 0)\}$ is linearly independent or not?

Solution:

Given $S = \{(1, 0, 0), (0, 1, 0), (1, 1, 1), (1, 1, 0)\}$

Let a, b, c, d be scalars. Then

$$a(1, 0, 0) + b(0, 1, 0) + c(1, 1, 1) + d(1, 1, 0) = (0, 0, 0)$$

$$(a + c + d, b + c + d, c) = (0, 0, 0)$$

$$\Rightarrow a + c + d = 0$$

$$b + c + d = 0$$

$$c = 0$$

$$\therefore c = 0, a + d = 0, b + d = 0$$

$$\Rightarrow a = b = -d$$

Hence, S is linearly dependent.

Remark

- The empty set is linearly independent
- The linearly dependent set must be nonempty
- A set consisting of a single nonzero vector is linearly independent
- $\{0\}$ linearly dependent
- Any subset of a vector space that contains $\{0\}$ is Linearly dependent

Theorem

Let V be a vector space and $S_1 \subseteq S_2 \subseteq V$. If S_1 is linearly dependent, then S_2 is linearly dependent.

Proof:

Given $S_1 \subseteq S_2 \subseteq V$ and S_1 is linearly dependent.

To prove: S_2 is linearly dependent.

Let $u_1, u_2, \dots, u_m \in S_1$. Then $a_1u_1 + a_2u_2 + \dots + a_mu_m = 0$.

Since S_1 is linearly dependent, atleast one scalar is non zero.
i.e., $a_i \neq 0$ for some i

Also, $a_i \neq 0 \in S_2$ for some i ($\because S_1 \subseteq S_2$)

i.e., For $u_1, u_2, \dots, u_m, u_{m+1}, \dots, u_n \in S_2$, we have

$a_1u_1 + a_2u_2 + \dots + a_mu_m + a_{m+1}u_{m+1} + \dots + a_nu_n = 0, a_i \neq 0$ for some i

$\Rightarrow S_2$ is linearly dependent.

Hence proved.

Corollary

Let V be a vector space and let $S_1 \subseteq S_2 \subseteq V$. If S_2 is linearly independent, then S_1 is linearly independent.

Theorem

Let S be a linearly independent subset of a vector space V and let v be a vector in V that is not in S . Then $S \cup \{v\}$ is linearly dependent if and only if $v \in \text{span}(S)$.

Proof:

Given S is linearly independent set of V and $v \in V$.

To prove: $S \cup \{v\}$ is linearly dependent $\Leftrightarrow v \in \text{span}(S)$.

Assume $S \cup \{v\}$ is linearly dependent.

To prove $v \in \text{span}(S)$.

Since $S \cup \{v\}$ is linearly dependent, we have

$$a_1 u_1 + a_2 u_2 + \cdots + a_n u_n = 0$$

for the vectors $u_1, u_2, \cdots, u_n \in S \cup \{v\}$ and some nonzero scalars $a_1, a_2, \cdots, a_n$.

Because S is linearly independent, one of the u_i 's (say u_1) equals v .
Thus,

$$a_1v + a_2u_2 + \cdots + a_nu_n = 0$$

$$\Rightarrow a_1v = -a_2u_2 - a_3u_3 - \cdots - a_nu_n$$

$$v = a_1^{-1}(-a_2u_2 - a_3u_3 - \cdots - a_nu_n)$$

$$= -a_1^{-1}a_2u_2 - a_1^{-1}a_3u_3 - \cdots - a_1^{-1}a_nu_n$$

$$= (-a_1^{-1}a_2)u_2 + (-a_1^{-1}a_3)u_3 + \cdots + (-a_1^{-1}a_n)u_n$$

$$\Rightarrow v \in \text{span}(S)$$

($\because v$ is a linear combination of vectors of S)

Conversely,

Let $v \in \text{span}(S)$.

To prove $S \cup \{v\}$ is linearly dependent.

Let $v \in \text{span}(S)$.

Then there exist vectors $v_1, v_2, \dots, v_m \in S$ and scalars $b_1, b_2, \dots, b_m$ such that $v = b_1v_1 + b_2v_2 + \dots + b_mv_m$

$$\Rightarrow 0 = b_1v_1 + b_2v_2 + \dots + b_mv_m - v$$

$$0 = b_1v_1 + b_2v_2 + \dots + b_mv_m + (-1)v$$

Since $v \neq v_i$ for $i = 1, 2, \dots, m$ the coefficient of v in this linear combination is nonzero, so the set $\{v_1, v_2, \dots, v_m, v\}$ is linearly dependent.

We know that "Let V be a vector space and $S_1 \subseteq S_2 \subseteq V$. If S_1 is linearly dependent, then S_2 is linearly dependent".

$\therefore S \cup \{v\}$ is linearly dependent.

Hence proved.

BASIS AND DIMENSIONS

Basis

A **basis** β for a vector space V is a linearly independent subset of V that generates V .

i.e., β is a basis for V if

- (i) $\text{span}(\beta) = V$ or $L(\beta) = V$
- (ii) β is linearly independent

If β is a basis for V , we also say that the vectors of β form a basis for V .

Example 1

Determine the set $S = \{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ is a basis for R^3 or not?

Solution: Given $S = (1, 1, 0), (1, 0, 1), (0, 1, 1)$

Let $a, b, c \in R$. Then

$$a(1, 1, 0) + b(1, 0, 1) + c(0, 1, 1) = (x, y, z)$$

$$(a, a, 0) + (b, 0, b) + (0, c, c) = (x, y, z)$$

$$(a + b, a + c, b + c) = (x, y, z)$$

$$\Rightarrow a + b = x$$

$$a + c = y$$

$$b + c = z$$

Solving the above equations, we get

$$a = \frac{1}{2}(x + y - z); b = \frac{1}{2}(x - y + z); c = \frac{1}{2}(-x + y + z)$$

$$(x, y, z) = \frac{1}{2}(x+y-z)(1, 1, 0) + \frac{1}{2}(x-y+z)(1, 0, 1) + \frac{1}{2}(-x+y+z)(0, 1, 1)$$

Hence every vector in R^3 is spanned by S . i.e., $R^3 = \text{Span}(S)$

Let a, b, c be scalars. Then

$$a(1, 1, 0) + b(1, 0, 1) + c(0, 1, 1) = (0, 0, 0)$$

$$(a + b, a + c, b + c) = (0, 0, 0)$$

$$\Rightarrow a + b = 0$$

$$a + c = 0$$

$$b + c = 0$$

$$\Delta = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} = -1 - 1 + 0 = -2 \neq 0$$

$\therefore$ the set S is linearly independent.

Hence the set $S = \{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ is a basis of R^3

Example 2

Determine the set $S = \{(2, -4, 1), (0, 3, -1), (6, 0, -1)\}$ is a basis for R^3 or not?

Solution: Given $S = \{(2, -4, 1), (0, 3, -1), (6, 0, -1)\}$. Then

$$a(2, -4, 1) + b(0, 3, -1) + c(6, 0, -1) = (0, 0, 0)$$

$$(2a + 6c, -4a + 3b, a - b - c) = (0, 0, 0)$$

$$\Rightarrow 2a + 6c = 0$$

$$-4a + 3b = 0$$

$$a - b - c = 0$$

$$\Delta = \begin{vmatrix} 2 & 0 & 6 \\ -4 & 3 & 0 \\ 1 & -1 & -1 \end{vmatrix} = 2(-3) + 6(1) = 0$$

$\therefore$ the vectors are linearly dependent.

Hence S won't form a basis of R^3 .

Theorem

Let V be a vector space and $\beta = \{u_1, u_2, \dots, u_n\}$ be a subset of V . Then β is a basis for V if and only if each $v \in V$ can be uniquely expressed as a linear combination of vectors of β , i.e., can be expressed in the form $v = a_1u_1 + a_2u_2 + \dots + a_nu_n$ for unique scalars $a_1, a_2, \dots, a_n$.

Proof:

Given: V be a vector space

$\beta = \{u_1, u_2, \dots, u_n\}$ be a subset of V

To prove: β is a basis for $V \Leftrightarrow v \in V$ can be uniquely expressed as a linear combination of vectors of β .

Assume β is a basis for V .

If $v \in V$, then $v \in \text{span}(\beta)$ ($\because \text{span}(\beta) = V$)

Thus, v is a linear combination of the vectors of β .

Suppose that

$$v = a_1u_1 + a_2u_2 + \cdots + a_nu_n$$

$v = b_1u_1 + b_2u_2 + \cdots + b_nu_n$ are two representations of v .

Then

$$v - v = (a_1u_1 + a_2u_2 + \cdots + a_nu_n) - (b_1u_1 + b_2u_2 + \cdots + b_nu_n)$$

$$\Rightarrow (a_1 - b_1)u_1 + (a_2 - b_2)u_2 + \cdots + (a_n - b_n)u_n = 0$$

$$\Rightarrow a_1 - b_1 = 0, a_2 - b_2 = 0, \cdots, a_n - b_n = 0 \quad (\because \beta \text{ is L.I})$$

$$\Rightarrow a_1 = b_1, a_2 = b_2, \cdots, a_n = b_n$$

$\Rightarrow V$ can be uniquely expressible as a linear combination of the vectors of β .

Conversely,

Assume each $v \in V$ can be uniquely expressed as a linear combination of vectors in β

To prove: β is a basis of V .

Since each $v \in V$ can be uniquely expressed as a linear combination of vectors in β , we have $\text{span}(\beta) = V$

For any $u_1, u_2, \dots, u_n \in \beta$, we have $a_1u_1 + a_2u_2 + \dots + a_nu_n = 0$

Also $0u_1 + 0u_2 + \dots + 0u_n = 0$

But given $v \in V$ can be expressed uniquely.

$$\Rightarrow a_1u_1 + a_2u_2 + \dots + a_nu_n = 0u_1 + 0u_2 + \dots + 0u_n$$

$$\Rightarrow a_1 = 0, a_2 = 0, \dots, a_n = 0$$

$\therefore \beta$ is linearly independent.

Hence β is a basis of V .

Theorem

If a vector space V is generated by a finite set S , then some subset of S is a basis for V .

Hence V has a finite basis.

Proof:

Given $\text{span}(S) = V$.

To prove: Subset of S is a basis for V

If $S = \emptyset$ or $S = \{0\}$, then $V = \{0\}$ and $\emptyset$ is a subset of S that is a basis for V .

Otherwise S contains a nonzero vector u_1 .

We know that "A set consisting of a single non zero vector is linearly independent".

$\therefore S = \{u_1\}$ is linearly independent

Continue, if possible, choosing vectors $u_2, \dots, u_k \in S$ such that $\{u_1, u_2, \dots, u_k\}$ is linearly independent.

Since S is a finite set, we must eventually reach a stage at which $\beta = \{u_1, u_2, \dots, u_k\}$ is a linearly independent subset of S , but adjoining to β any vector in S not in β produces a linearly dependent set.

We have to prove that β is a basis for V .

Because β is linearly independent by construction, it suffices to show that β spans V . i.e., $\text{span}(\beta) = V$

Given $\text{span}(S) = V$.

So it is enough to prove $\text{span}(S) = \text{span}(\beta)$

By Theorem "*The span of any subset S of a vector space V is a subspace of V . Moreover, any subspace of V that contains S must also contain the span of S* "

From the above fact, we have $\text{span}(\beta) \subseteq S$ — — — — — (1)

Now, we need to prove $S \subseteq \text{span}(\beta)$.

Let $v \in S$.

If $v \in \beta$, then clearly $v \in \text{span}(\beta)$.

If $v \notin \beta$, then $\beta \cup \{v\}$ is linearly dependent.

We know that "Let S be a linearly independent subset of a vector space V and let $v \in V$ that is not in S . Then $S \cup v$ is linearly dependent $\Leftrightarrow v \in \text{span}(S)$ ".

From the above fact, we have $v \in \text{span}(S)$

$\Rightarrow S \subseteq \text{span}(\beta) \text{ --- (2)}$.

From (1) and (2), we have $\text{span}(S) = \text{span}(\beta)$

$\therefore V = \text{span}(\beta)$. i.e., β generates V

Hence β is a basis of V

Hence proved.

Replacement Theorem

Let V be a vector space that is generated by a set G containing exactly n vectors and let L be a linearly independent subset of V containing exactly m vectors. Then $m \leq n$ and there exists a subset H of G containing exactly $n - m$ vectors such that $L \cup H$ generates V .

Corollary 1

Let V be a vector space having a finite basis. Then every basis for V contains the same number of vectors.

Dimension

The number of vectors in each basis for V is called the **dimension** of V and is denoted by $\dim(V)$.

Finite dimensional vector space

A vector space V is called **finite dimensional** if it has a basis consisting of a finite number of vectors.

ie., $\dim(V) = \text{Finite no. of elements}$

Infinite dimensional vector space

A vector space that is not finite-dimensional is called **infinite-dimensional vector space**.

S.No	Set	Standard Basis	Dimension
1	$V = \{0\}$	$S = \phi$	0
2	$V = R(R)$	$S = \{1\}$	1
3	$V = C(C)$	$S = \{1\}$	1
4	$V = C(R)$	$S = \{1, i\}$	2
5	$V = F^n$	$S = \{e_1, e_2, \dots, e_n\}$	n
6	$V = P_n(F)$	$S = \{1, x, x^2, \dots, x^n\}$	$n + 1$
7	$V = P(F)$	$S = \{1, x, x^2, \dots\}$	<i>infinite</i>
8	$V = M_{(m \times n)}(F)$	$S = \{E^{ij} / 1 \leq i \leq m, 1 \leq j \leq n\}$ where, E^{ij} denote the matrix whose only nonzero entry is 1 in the i^{th} row and j^{th} column.	mn

Corollary 2

Let V be a vector space with dimension n .

- a. Any finite generating set for V contains at least n vectors and a generating set for V that contains exactly n vectors is a basis for V .
- b. Any linearly independent subset of V that contains exactly n vectors is a basis for V .
- c. Every linearly independent subset of V can be extended to a basis for V .

The Dimension of Subspaces

Example

- ① Let $V = F^5$ and $W = \{(a_1, a_2, a_3, a_4, a_5) \in F^5 / a_1 + a_3 + a_5 = 0, a_2 = a_4\}$. Then
Basis of W : $S = \{(-1, 0, 1, 0, 0), (-1, 0, 0, 0, 1), (0, 1, 0, 1, 0)\}$.
Dimension: $\dim(W) = 3$.
- ② Let $V = M_{(n \times n)}(F)$ and $W = \{A_{(n \times n)} / A \text{ is diagonal matrix}\}$. Then
Basis of W : $S = \{E^{11}, E^{22}, \dots, E^{nn}\}$, where E^{ij} is the matrix in which the only nonzero entry is 1 in the i th row and j th column.
Dimension: $\dim(W) = n$
- ③ Let $V = M_{(n \times n)}(F)$ and $W = \{A_{(n \times n)} / A = A^T\}$. Then
Basis of W : $S = \{E^{ij} / 1 \leq i \leq j \leq n\}$, where A^{ij} is the $(n \times n)$ matrix having 1 in the i th row and j th column and 1 in the j th row and i th column, otherwise zero.
Dimension: $\dim(W) = n + (n - 1) \cdots + 1 = \frac{1}{2}n(n + 1)$

Theorem

Let W be a subspace of a finite-dimensional vector space V . Then W is finite-dimensional and $\dim(W) \leq \dim(V)$.
Moreover, if $\dim(W) = \dim(V)$, then $V = W$.

Proof:

Let $\dim(V) = n$.

If $W = \{0\}$, then W is finite-dimensional.

$$\begin{aligned}\therefore \dim(W) &= 0 \leq n \\ \Rightarrow \dim(W) &\leq \dim(V).\end{aligned}$$

Otherwise, W contains a nonzero vector x_1

$\Rightarrow \{x_1\}$ is a linearly independent set.

Continue choosing vectors, $x_1, x_2, \dots, x_k \in W$ such that $\{x_1, x_2, \dots, x_k\}$ is linearly independent.

This process must stop at a stage where $k \leq n$

($\because$ no linearly independent subset of V can contain more than n vectors)

$\therefore \{x_1, x_2, \dots, x_k\}$ is linearly independent but adjoining any other vector from W produces a linearly dependent set.

We have to prove $\{x_1, x_2, \dots, x_k\}$ generates W . i.e.,
 $W = \text{span}(\{x_1, x_2, \dots, x_k\})$

By Theorem "*The span of any subset S of a vector space V is a subspace of V . Moreover, any subspace of V that contains S must also contain the span of S* "

By the above fact, we have $\text{span}(\{x_1, x_2, \dots, x_k\}) \subseteq W$

To prove $W \subseteq \text{span}(\{x_1, x_2, \dots, x_k\})$

Let $w \in W$. Then we have to prove $w \in \text{span}(\{x_1, x_2, \dots, x_k\})$

Suppose $w \notin \text{span}(\{x_1, x_2, \dots, x_k\})$, then $\{x_1, x_2, \dots, x_k, w\}$ is linearly independent.

Which is the contradiction.

$\therefore w \in \text{span}(\{x_1, x_2, \dots, x_k\})$

$\therefore W \subseteq \text{span}(\{x_1, x_2, \dots, x_k\})$

Hence $\{x_1, x_2, \dots, x_k\}$ generates W

$\therefore \{x_1, x_2, \dots, x_k\}$ is a basis for W .

Hence $\dim(W) = k \leq n \Rightarrow \dim(W) \leq \dim(V)$.

If $\dim(W) = n$, then a basis for W is a linearly independent subset of V containing n vectors.

We know that "*Let V be a vector space with dimension n . Then any linearly independent subset of V that contains exactly n vectors is a basis for V* ".

By the above fact W is also a basis for V

Hence $W = V$.

Corollary

If W is a subspace of a finite-dimensional vector space V , then any basis for W can be extended to a basis for V .

Minimal generating set

Let V be a vector space over the field F and S contained in V , then S is called **minimal generating set** for V if

- (i) $L(S) = V$
- (ii) no proper subset of S will generate V .

Maximal linearly independent subsets

Let S be a subset of a vector space V , a **maximal linearly independent subset** of S , the subset B of S satisfying the following conditions,

- (i) B is linearly independent.
- (ii) The only linearly independent subsets of S that contain B is B itself.